

CSE 464 Project Part 3: Design Patterns & Refactoring

Author: Sai Mahesh Nomula (snomula)

GitHub Repository: <https://github.com/Sai-2811/CSE464-2026-smahesh> **Pull Request Link:** <https://github.com/Sai-2811/CSE464-2026-smahesh/pull/1>

Hi! This is my submission for Part 3 of the CSE 464 course project. For this assignment, I applied code refactoring principles, introduced the Template Method and Strategy design patterns, and implemented a Random Walk search algorithm.

My Environment Setup

- **Java:** JDK 11 (or higher)
- **Build Tool:** Maven 3.8+
- **IDE:** IntelliJ IDEA Community Edition

Feature 1: Refactoring (5 Commits)

I applied 5 separate refactorings to improve the structure and readability of the Graph application.

- **Refactor #1 (Magic Strings to Constants):** [baf63c0](#)
- **Refactor #2 (Extract Methods in Parser):** [fd0ac81](#)
- **Refactor #3 (Extract Methods in Exporter):** [1262a39](#)
- **Refactor #4 (Extract toString Helpers):** [fdde00b](#)
- **Refactor #5 (Introduce Node & Update Path):** [a48906e](#)

Feature 2: Template Method & Strategy Pattern

I applied design patterns to abstract the graph searching logic.

- **Apply Template Method Pattern (BFS/DFS):** [2285f51](#)
- **Apply Strategy Pattern (Factory + Interface):** [0f21b18](#)

Feature 3: Random Walk Algorithm

Using the new design patterns, I added a Random Walk search algorithm.

- **Add Random Walk algorithm:** [b4aed30](#)

Feature 4: Pull Request & Peer Review

I opened a Pull Request for the `refactor` branch and completed a 5-comment code review, which I subsequently addressed via micro-commits.

- **Address Review Comment #1 (Extract helper):** [da3c2e9](#)
- **Address Review Comment #2 (Javadoc for factory):** [9769398](#)
- **Address Review Comment #3 (Clarify constructor):** [c84f3ab](#)
- **Address Review Comment #4 (Document immutability):** [fb2d69b](#)
- **Address Review Comment #5 (Explain test1.dot):** [6e319f3](#)

Thanks for reviewing my project!